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#include <iostream>
#include <string>
#include <algorithm>

using namespace std;

struct Student {
   int id;
   string name;
   float cgpa;
};

// 1. Bubble Sort by Name
void sortByName(Student arr[], int n) {
   for (int i = 0; i < n - 1; i++) {
      for (int j = 0; j < n - i - 1; j++) {
         if (arr[j].name > arr[j + 1].name) {
            swap(arr[j], arr[j + 1]);
         }
      }
   }
}

// 2. Selection Sort by CGPA (Descending)
void sortByCGPA(Student arr[], int n) {
   for (int i = 0; i < n - 1; i++) {
      int maxIdx = i;
      for (int j = i + 1; j < n; j++) {
         if (arr[j].cgpa > arr[maxIdx].cgpa) maxIdx = j;
      }
      swap(arr[i], arr[maxIdx]);
   }
}

// 3. Linear Search
int linearSearch(Student arr[], int n, int targetId) {
   for (int i = 0; i < n; i++) {
      if (arr[i].id == targetId) return i;
   }
   return -1;
}

int main() {
   int n = 3;
   // Using an array of size 3 for simple demonstration
   Student db[3] = {
      {103, "Alice", 3.8},
      {101, "Charlie", 3.5},
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        {102, "Bob", 3.9}
    };

    cout << "--- Initial Database ---\n";
    for(int i=0; i<n; i++) cout << db[i].id << " | " << db[i].name << " | " << db[i].cgpa << "\n";

    // Sort and Display by Name
    sortByName(db, n);
    cout << "\n--- Sorted Alphabetically by Name ---\n";
    for(int i=0; i<n; i++) cout << db[i].name << " (ID: " << db[i].id << ")\n";

    // Sort and Display by CGPA
    sortByCGPA(db, n);
    cout << "\n--- Sorted by Highest CGPA ---\n";
    for(int i=0; i<n; i++) cout << db[i].name << " : " << db[i].cgpa << "\n";

    // Search Demonstration
    int target = 102;
    int index = linearSearch(db, n, target);
    cout << "\n--- Search Result for ID " << target << " ---\n";
    if(index != -1) cout << "Found! Name: " << db[index].name << ", CGPA: " << db[index].cgpa << "\n";
    else cout << "Not Found\n";

    return 0;
}
```